

LLM Usage Statement

Jincheng Yang

I used ChatGPT as a learning and debugging assistant while completing this assignment. ChatGPT helped explain concepts, review errors, suggest code changes, and organize parts of the report. I reviewed the explanations, ran the code, checked the results, and decided what to include in the final submission.

Model Used:

ChatGPT (OpenAI GPT-5).Part A Question 2: Item Response Theory

“Please explain the IRT model, log-likelihood, and gradients for theta and beta step by step.”

“Explain the sigmoid, evaluate, and np.add.at functions in item_response.py.”

“Help me understand and fix the errors in my IRT code.”

“Explain the differences between training, validation, and test data.”

“Help me create and interpret the log-likelihood and question-response plots.”Part A Question 4: Ensemble

“Explain bootstrap sampling and how to combine three IRT models using bagging.”

“Guide me through ensemble.py and help me fix the indentation error.”

“Help me compare the individual models and ensemble results and explain why the ensemble did not always perform better.”

Part B: Modified IRT

“Explain the requirements for Part B and help me choose an extension to the original IRT model.”

“Explain the 2PL IRT model and the new discrimination parameter alpha.”

“Explain why gamma is initialized to zero and why alpha is defined as $\exp(\gamma)$.”

“Help me derive and understand the gradients for theta, beta, and gamma.”

“Help me add L2 regularization and design experiments using the validation set.”

“Help me organize the results, figures, comparison, and limitations for the Part B report.”

ChatGPT outputs are not included, as required by the assignment instructions.

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The large language model I used during the completion of this project is ChatGPT. The prompts are shown below:

Explain the architecture and training process of the neural network model, including the roles of the latent dimension, learning rate, number of epochs, and L2 regularization.

Debugging: help to find implementation issues, including missing variables (train_objectives, valid_objectives), incorrect function calls, and evaluation logic.

Explain the effects of different hyperparameters (e.g., latent dimension, learning rate, regularization)

How to select appropriate hyperparameters based on validation performance.

Explain the effects of different hyperparameters (e.g., latent dimension, learning rate, regularization).